

99_appendix

Appendix A — Global Troubleshooting

Symptom	Likely cause	Fix
<code>dbt debug</code> fails to connect	wrong project/dataset/location or no ADC	set <code>location: US</code> (the GA4 sample is US), run <code>gcloud auth application-default login</code> , check <code>profiles.yml</code> project.
"Dataset not found: ga4_obfuscated_sample_ecommerce"	location mismatch	the public dataset is in US ; your target dataset + job location must be US.
Query scans 100s of GB / huge bill	no shard pruning / <code>SELECT *</code>	always filter <code>_TABLE_SUFFIX BETWEEN ...</code> ; set <code>maximum_bytes_billed</code> ; never <code>SELECT * over events_*</code> .
Step rate > 1 (e.g. <code>view_to_cart_rate = 1.4</code>)	occurrence-based flags, not monotonic	use <code>reached_*</code> max-downstream flags (M3); rerun the monotonicity test.
<code>dim_channels</code> test fails	an 11th channel group crept in	keep exactly 10 groups; channel logic only in <code>channel_group_case()</code> .
Revenue / AOV ~2× too high	<code>transaction_id</code> not deduped	dedup orders by <code>transaction_id</code> in <code>fct_orders</code> (M4).
<code>revenue</code> ≠ <code>gross_revenue</code> totals	session-attributed vs order-grain	they reconcile only at the grand total ; don't compare per-slice.
<code>validate_semantic.py</code> fails (dangling ref)	a metric's numerator/denominator/expr names a metric not in the registry	fix the reference or add the metric; the registry is the single source of truth.
<code>semantic-mcp</code> can't find the registry	filename drift	<code>mcp_servers.yml</code> must point at <code>models/semantic/semantic_layer.yml</code> .
<code>run_query</code> raises <code>NotDryRunFirst</code>	called without a prior <code>dry_run</code>	always <code>dry_run</code> first (this is the guardrail working — don't bypass it).
<code>decompose_change</code> golden test off	wrong mix/rate/interaction algebra	$\text{mix} = \sum \Delta w_i \cdot r_i(t_0)$, $\text{rate} = \sum w_i(t_0) \cdot \Delta r_i$, $\text{interaction} = \sum \Delta w_i \cdot \Delta r_i$; seed <code>rng_seed=1729</code> .
Eval accuracy looks great but agents hallucinate	grading on live data, no hallucination gate	grade against the injected labels ; make hallucination a hard-zero CI gate.

Symptom	Likely cause	Fix
Critic flags everything seasonal as real	<code>seasonality_calendar</code> not seeded	seed Black Friday 2020 / Dec peak / Jan trough (M8).
Run never finishes / huge token cost	row-level dumps into LLM context	summarize query results to aggregates before they enter the model; enforce per-agent tool allow-lists.
LLM output not byte-reproducible	expecting determinism from the model	the agent layer is graded statistically by the eval harness, not asserted byte-identical; only stats/SQL are deterministic.

Appendix B — Definition of Done (per level)

- **L1 (after M7):** governed marts + semantic layer + the minimal loop; `reconcile('revenue', 'day')` matches a hand-written control query to the cent; **0 hallucinated columns**; one anomaly → a Decision Brief in **<5 min**.
- **L2 (after M10):** all 7 agents + 5 MCP servers + the Critic + the eval harness in CI; every finding carries significance + dollar impact + an experiment; **root-cause ≥85% vs ≤45% baseline**; cost under the byte budget.
- **L3 (after M11):** autonomous scheduled runs with memory-driven learning; forecasting / cohorts / RFM; full Critic refutation battery; accuracy holds across all canonical dimensions.

Appendix C — Minimum-viable Helios (if you have limited time)

You do not have to build all of M0–M12 to have something valuable:

- **A weekend (M0–M5):** the governed dbt marts + the validated semantic layer. This alone is a strong analytics-engineering portfolio piece — correct, tested, documented metrics on real GA4 data, queryable by anyone. **No LLM required.**
- **A week (+ M6, M6b, M7):** add the MCP servers and the minimal loop → governed SQL + deterministic stats + a single automated Decision Brief. This is the "grounded AI analyst" story end-to-end on one path.
- **Two–three weeks (+ M8–M10):** the full 7-agent loop and the benchmark → the headline **85%-vs-45%** result, which is the project's central, defensible claim. This is the L2 portfolio centerpiece.

Build depth-first along the critical path (`DEPENDENCY_MAP.md` §4), not breadth-first. Finance facts, cohorts, the scheduler, and the drill-down UI can all wait.

Appendix D — Build checklist

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[ ] M0  repo + GCP/IAM + dbt config      (dbt debug green)
[ ] M1  sources + macros + seed         (dbt deps/seed)
[ ] M2  staging                          (staging tests green)
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[ ] M3 sessionization + funnel KEYSTONE (monotonicity + uniqueness golden tests) ★ test first
[ ] M4 marts (reconcile to the cent; channels = 10)
[ ] M5 semantic layer live (validate_semantic.py → 0 dangling refs)
[ ] M6 semantic-mcp + warehouse-mcp (round-trip + budget gate)
[ ] M6b stats + experiment + report MCP (decompose_change golden test)
[ ] M7 minimal loop (anomaly → brief <5 min; 0 hallucination) ✓ L1
[ ] M8 memory (save/recall; calendars seeded)
[ ] M9 full 7-agent loop (PASS findings: significance + $ + experiment)
[ ] M10 eval + CI (≥85% vs ≤45%; hallucination 0) ✓ L2
[ ] M11 autonomy + depth (<5 min/run scheduled; all dims) ✓ L3
[ ] M12 productionization / frontier (deferred)

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Appendix E — Where the code lives (spec doc index)

DBT_GUIDE.md (all dbt code: config, sources, staging, intermediate, marts, tests, freshness, lineage, docs) ·
 DATA_MODEL.md (tables, grains, keys, ER) · models/semantic/semantic_layer.yaml (the registry) ·
 MCP_ARCHITECTURE.md (5 servers + skeletons) · AGENT_ARCHITECTURE.md (7 agents + FSM + Critic + RCA) ·
 eval/scenarios/scenarios.yaml (50 labeled scenarios) · METRIC_GOVERNANCE_GUIDE.md +
 METRIC_DEPENDENCY_GRAPH.md (metric governance) · DEPENDENCY_MAP.md + DEVELOPMENT_PLAN.md (build
 order) · CLAUDE.md (conventions + grounding rules) · HELIOS_PROJECT_BIBLE.md (the full 25-section
 reference).

If you build nothing else, build M0–M5 correctly. Everything downstream trusts those marts — and they fail *silently* if the keystones (M3) are wrong, so test them first and reconcile to the cent.